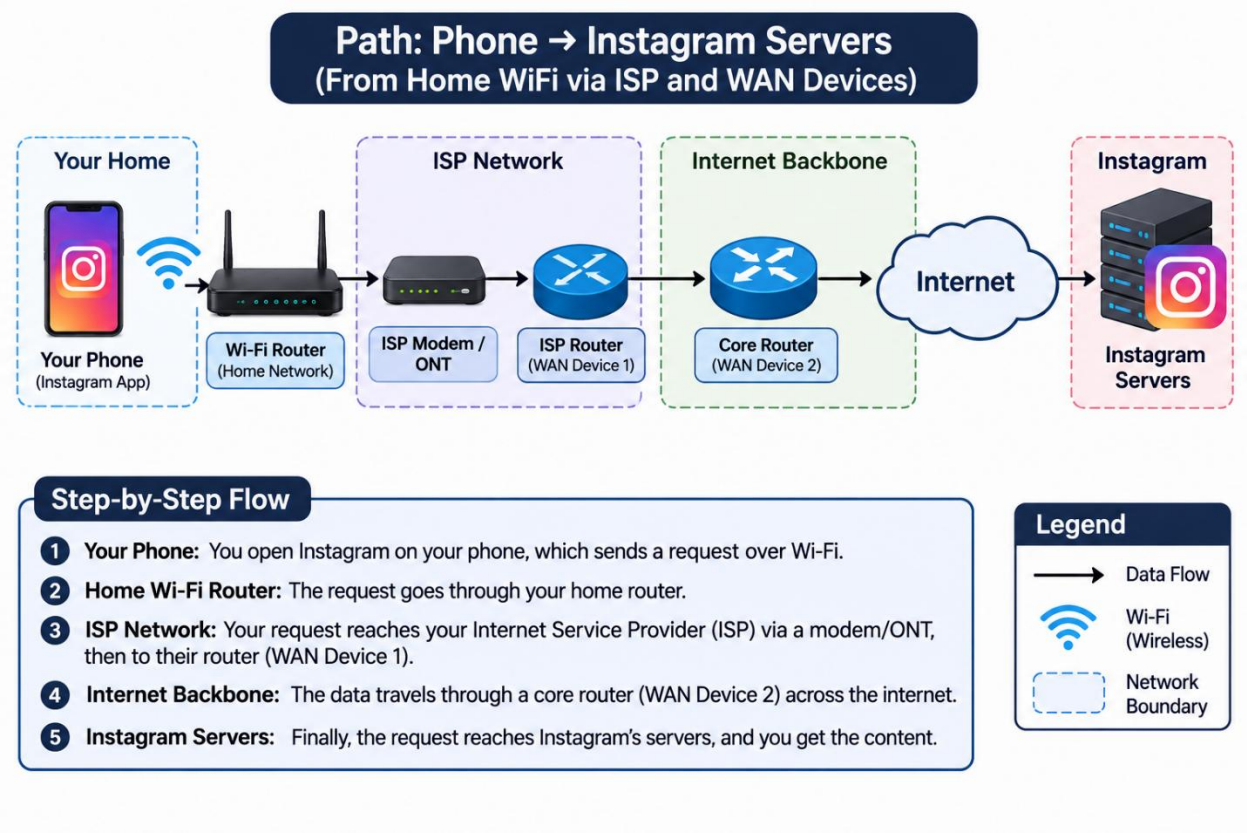
Wide Area Networks

Task 1 : Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.



To understand how a mobile phone connects to Instagram servers through a home Wi-Fi network and a Wide Area Network (WAN).

Description:

When a user opens Instagram on a mobile phone, the phone connects to the home Wi-Fi router. The router forwards the request to the Internet Service Provider (ISP). The request then travels through multiple WAN devices, such as ISP edge routers and core network routers, before reaching Instagram's servers.

Network Components:

1. Mobile Phone – Runs the Instagram application.
2. Home Wi-Fi Router – Connects the phone to the internet.

3. ISP Network – Provides internet connectivity.
4. ISP Edge Router – Forwards network traffic.
5. Core Network Router – Routes data across the WAN.
6. Instagram Servers – Process user requests and deliver content.

Conclusion:
Successfully illustrated how a mobile phone communicates with Instagram servers through a home Wi-Fi network, ISP infrastructure, and WAN devices.

Task 2 : Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

To compare three common WAN technologies based on speed, cost, and real-world applications.

WAN Technology	Speed	Cost	Real-World Usage
MPLS	High and predictable	High	Connecting different Zomato office locations.
Leased Line	High and dedicated	Very High	Connecting important IRCTC data centres and offices.
Broadband Internet	Medium to High	Low	Home internet connection used for browsing, streaming and online work.

Description:

1. **MPLS:** MPLS is used by organizations to connect multiple office locations through a managed network. It provides predictable traffic forwarding.
2. **Leased Line:** A leased line provides a dedicated connection between two locations with consistent bandwidth and reliable performance.
3. **Broadband Internet:** Broadband provides internet connectivity at an affordable cost and is commonly used in homes and small businesses.

Conclusion:

MPLS, Leased Line, and Broadband Internet are three important WAN technologies. Their selection depends on the organization's requirements, budget, speed, and reliability.

Task 3 : Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.

To understand how a Virtual Private Network (VPN) uses a WAN to provide secure internet access over public Wi-Fi.

What is VPN?

A Virtual Private Network (VPN) is a technology that creates an encrypted connection between a user's device and a VPN server over the internet.

How VPN Works:

1. A user connects their mobile phone or laptop to public Wi-Fi at a coffee shop.
2. The user opens a VPN application and connects to a VPN server.
3. The VPN creates an encrypted private tunnel over the public WAN.
4. Internet traffic travels through this encrypted tunnel to the VPN server.
5. The VPN server forwards the request to applications like YouTube or Spotify.
6. The response returns through the VPN connection to the user's device.

Benefits of VPN:

- Encrypts internet traffic between the device and VPN server.
- Helps protect sensitive information on public Wi-Fi.
- Provides a more secure connection over public networks.
- Hides the user's original IP address from destination websites by showing the VPN server's IP address.

Conclusion:
A VPN uses a public WAN to establish an encrypted private tunnel between a user's device and a VPN server. It helps protect internet communication when using public Wi-Fi networks.

Task 4 : Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

To select a suitable WAN technology for connecting Flipkart warehouses located in Ahmedabad, Mumbai, and Delhi.

Recommended WAN Technology:
MPLS (Multiprotocol Label Switching).

Description:
Flipkart needs a reliable network to connect its warehouses in different cities. MPLS is a suitable WAN technology because it allows multiple locations to communicate through a managed network.

Reasons for Selecting MPLS:

1. **Reliable Connectivity:** MPLS provides predictable network performance and helps warehouses exchange important information such as inventory, orders, and shipment details.
2. **Multiple Location Connectivity:** MPLS allows Ahmedabad, Mumbai, and Delhi warehouses to communicate through a single managed WAN infrastructure.

Conclusion:
MPLS is a suitable WAN technology for connecting Flipkart warehouses because it provides reliable communication and efficient connectivity between multiple locations.